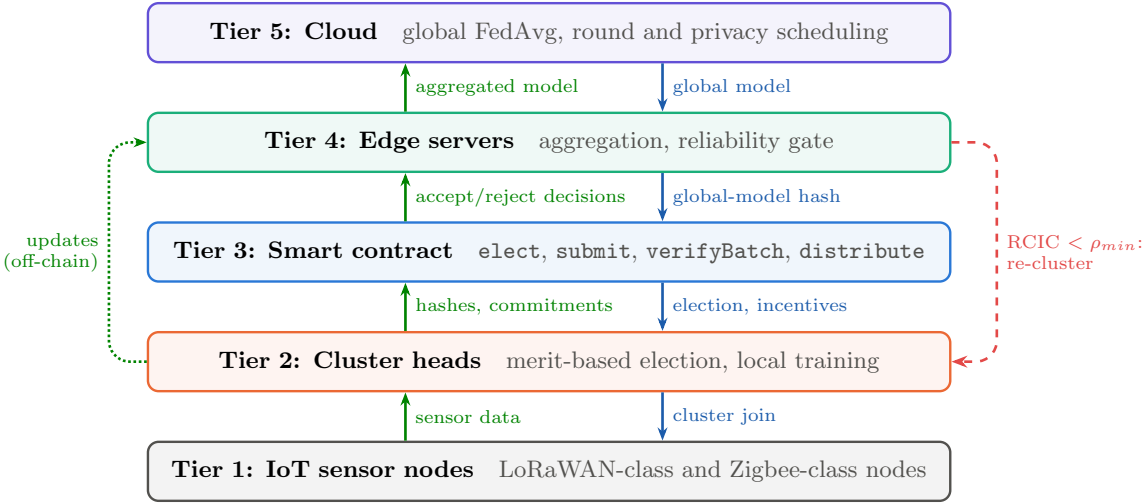

Tier 5: Cloud global FedAvg, round and privacy scheduling
aggregated model
global model
Tier 4: Edge servers aggregation, reliability gate
accept/reject decisions
global-model hash
Tier 3: Smart contract elect, submit, verifyBatch, distribute
hashes, commitments
election, incentives
Tier 2: Cluster heads merit-based election, local training
sensor data
cluster join
Tier 1: IoT sensor nodes LoRaWAN-class and Zigbee-class nodes
updates (off-chain)
RCIC < ρ_{min}: re-cluster